1. Problem Statement

Item	Description / Details	Status
Problem	Travelers in South Africa struggle to find and book affordable, trusted hotels in one place.	Functional
Existing challenges	Some platforms charge high commissions, require international credit cards, and do not show local prices in South African Rand (R).	Functional
Example hotel information	Monastero Santa Rosa Hotel & Spa; rating 4.96; 124 reviews; private villa; ocean view; infinity pool; breakfast included; free Wi-Fi; room service.	Functional
Required outcome	Allow users to view hotel information and book conveniently, with a local EFT payment option ("Pay with Banking").	Functional

2. Feature Statements

Item	Description / Details	Status
a) User Registration (Signup)	Creates an account using name, email, and password through Firebase Authentication 13.1.0.	Fully functional
b) User Login (Signin)	Signs in with email and password using signInWithEmailAndPassword. Fixed missing .value issue in email input. Network issue was traced to mixed SDK versions and school Wi-Fi blocking identitytoolkit.googleapis.com.	Fully functional
c) Explore Hotels Page	Lists hotels from Firestore with image, hotel name (Conca dei	Fully functional

	Marini), rating 4.96, and price R780 per night.	
d) Room Details Page	Displays hotel image, title, description, amenities, and price of R780 per night. Example description: a 17th-century Dominican monastery restored with terraced gardens, infinity pool overlooking the sea, and luxury spa.	Partial
e) Booking Form	Includes check-in (14 October), check-out (19 October), guest selector (2 guests), and total-price calculation.	Partial
f) Book Now – Pay with Banking	Sticky bottom button with gold gradient (#c89b5e), intended to trigger EFT payment.	Partial

3. Solution Statement

Item	Description / Details	Status
Platform	A web-based hotel booking platform built using Firebase 13.1.0 as Backend-as-a-Service (BaaS).	Functional
Configuration	config.js centralizes Firebase initialization (project ID: hotel-booking-2bb67).	Functional
Authentication	Firebase Authentication is used for user registration and login.	Functional
Data storage	Cloud Firestore stores users, hotels, and bookings.	Functional
Page navigation	URL parameters pass hotel information between pages.	Functional
Local payment	The platform aims to address local payment needs through “Pay with Banking” EFT.	Partial

4. Technology Statement

Technology	Details
Frontend	HTML5, CSS3, Vanilla JavaScript ES6 modules, and Poppins font.
Backend / SDK	Firebase JavaScript SDK 13.1.0 (firebase-app.js, firebase-auth.js, and firebase-firestore.js from gstatic.com).
Database	Cloud Firestore collections: users, hotels, and bookings.
Authentication	Firebase Email/Password authentication (identitytoolkit.googleapis.com) .
Testing tools	Visual Studio Code Live Server at 127.0.0.1:5500 and Chrome DevTools.
Icons and images	Font Awesome 6.5.0 and Unsplash images.

5. Design Statement

Item	Description / Details	Status
UI theme	Luxury hotel theme using a gold gradient (#c89b5e to #b0844a) for the login button.	
Cards and effects	White translucent cards at 92% opacity, backdrop blur, 16 px border radius, and soft shadows.	
Responsive layout	Mobile-first responsive design.	
Room gallery	Main image approximately 400 px; thumbnails 90 × 70 px with an active gold border.	
User flow	signin.html → explore.html → room-details.html → booking confirmation.	

Usability	A sticky "Book Now" button is intended to improve conversion and make booking easier.	

6. Testing Statement – PA1406 Test and Debug

Test / Item	Procedure / Result	Outcome
Test environment	Chrome; Live Server at 127.0.0.1:5500; Firebase 13.1.0; school Wi-Fi and mobile hotspot.	
Test Case 1 – Valid login	Input: otlotlengsepamo905@gmail.com. Expected: redirect to explore.html. First run returned auth/network-request-failed at signin.js:61 because of mixed SDK versions (11.1.0 vs 13.1.0) and network/firewall blocking. After unifying versions to 13.1.0 and using a hotspot, login succeeded.	Fully functional
Test Case 2 – Missing .value bug	Using getElementById('email') without .value caused auth/invalid-email. Fixed by using emailInput.value.trim() and passwordInput.value.	Fully functional
Test Case 3 – Invalid password	auth/wrong-password is displayed for an incorrect password.	Fully functional
Test Case 4 – Gallery main image	changeImg(this) switches the main image.	Fully functional
Test Case 5 – Different pictures	Thumbnail onclick updates mainImg.	Fully functional
Overall status	Core authentication and Explore page work fully. Room details, booking, and payment are partially functional.	Partial

7. Overall Project Summary

Component	Summary
Core authentication + Explore	Fully functional
Room details + Booking + Payment	Partial
Readiness	Partial